

Aggregate Fourier Defects in Rank-Growing Cloud Bridges

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July 2026

Abstract

The canonical pole counterloop is a rank-growing root-of-unity shell. We show that pointwise radial and phase convergence of its roots is not enough to transport trace moments. Fixed moments are controlled by aggregate low-frequency Fourier defects. A common phase shift tending to zero as $(N+1)^{-1/2}$ makes the third moment grow like $\sqrt{N}$, giving an exact counterexample to any maximum-phase-only bridge. The example is logical and is not asserted to describe the noisy cloud.

1 Aggregate criterion

Let $\theta_{N,j} = j\pi/(N+1)$ and consider conjugate pairs $r_{N,j}e^{\pm i(\theta_{N,j} + \varepsilon_{N,j})}$. Put $r_{N,j} = \beta + \delta_{N,j}$, and let $S_{N,n}^*$ denote the moment with $r_{N,j} = \beta$ and $\varepsilon_{N,j} = 0$.

Theorem 1.1 (fixed-moment stability). *For each fixed n , if $0 \leq r_{N,j}, \beta \leq B$ uniformly, then*

$$|S_{N,n} - S_{N,n}^*| \leq 2nB^n \sum_{j=1}^N |\varepsilon_{N,j}| + 2nB^{n-1} \sum_{j=1}^N |\delta_{N,j}|.$$

Consequently aggregate ℓ^1 phase and radial defects tending to zero imply the RH-272 moment bridge at every fixed order.

Proof. Use $|e^{in\varepsilon} - 1| \leq n|\varepsilon|$ and the mean-value bound $|r^n - \beta^n| \leq nB^{n-1}|r - \beta|$, then sum both conjugate members. $\square$

2 Maximum-phase counterexample

Take the exact radii $r_{N,j} = \beta$ and the common shift $\varepsilon_{N,j} = \Delta_N = (N+1)^{-1/2}$. The maximum phase error tends to zero. Since the unshifted third moment vanishes,

$$S_{N,3} = 2\beta^3 \sum_{j=1}^N \cos\left(\frac{3j\pi}{N+1} + 3\Delta_N\right).$$

The exact sine sum is $\sum_{j=1}^N \sin(3j\pi/(N+1)) = \cot(3\pi/(2(N+1)))$. Hence

Theorem 2.1 (common-shift divergence).

$$S_{N,3} = -\frac{4\beta^3}{\pi} \sqrt{N+1} + o(\sqrt{N}).$$

Thus $\max_j |\varepsilon_{N,j}| \rightarrow 0$ does not imply even a bounded third moment.

3 Boundary

The theorem invalidates a logical extrapolation from finite phase fits. It does not prove that the archived family has a coherent shift or fails the aggregate criterion. No Gate status changes.